

Total No. of Questions : 10]

SEAT No. :

P3849

[5561]-277

[Total No. of Pages : 2

B.E. (Computer Engineering)
SMART SYSTEM DESIGN AND APPLICATIONS
(2012 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data if necessary.*

- Q1)** a) What are various agent environments? Give PEAS representation for an agent. [8]
- b) Explain game theory and knowledge structure. [6]
- c) Explain the hardware requirements for robotics ? [4]

OR

- Q2)** a) Define problem formulation? Describe the components of problem with suitable example. [8]
- b) Explain rote learning with example. [6]
- c) Write short notes on Kalman Filters. [4]

- Q3)** a) Explain support Vector Machine with issues and applications. [4]
- b) Compare and contrast propositional logic and FOL. [6]
- c) What is Expert System? List out application of expert system? [4]

OR

- Q4)** a) Explain Role of NLP in AI. [4]
- b) What is baye's rule? State its application. [6]
- c) What is propositional logic? Explain with example. [4]

- Q5)** a) What is prior probability and posterior probability? Explain with suitable example. [6]
- b) Explain iterative deepening depth search algorithm with its function. [8]

OR

P.T.O.

- Q6)** a) What are the basic axioms of probability? Explain how to derive the useful facts from the basics axioms with suitable example. [8]
b) How to represent and evaluate decision problem with a decision network. [6]

- Q7)** a) What is supervised learning? Explain any one. [6]
b) Write a note on Robotics software architecture. [6]

OR

- Q8)** a) Explain in brief language models with suitable examples. [6]
b) Explain and draw a decision tree for deciding whether to wait for a table if a restaurant currently has no free tables. [6]

- Q9)** a) Explain in details the components that help in reconstructing the world in 3D. [6]
b) Enumerate and explain the different edge profiles using in edge detection. [6]

OR

- Q10)** a) What are the basic inference task that must be solved in a generic temporal model. [6]
b) List application domains of robotics. Explain any one in detail. [6]

